

**Your Time Zone or Mine? Geographic Configurations,
Global Team Coordination, and Project Outcomes**

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ABSTRACT

A team's configuration across distance and time can affect the coordination challenges its members need to overcome to get the job done. However, the effects of geographic distance and time separation are often confounded in global teams' research. In this paper we report on a survey study of 71 teams in a semiconductor manufacturing company to investigate how various distance and time configuration variables influence team coordination processes and outcomes. Our results show that the team's geographic configuration is associated with coordination processes, coordination outcomes and project performance and impact. Interestingly, these associations are observed with different geographic configuration variables. For example, the geographic distance only affects team communication and team knowledge. However, the number of time zones represented in a project is associated with longer communication delays, but it does not affect other process variables or coordination problems directly. But communication delay is positively associated with coordination problems. On the other hand, the maximum time zone spanned in a team is negatively associated with coordination problems, project performance and project impact. Teams with better task organization exhibited fewer coordination problems but, interestingly, none of the geographic configuration variables had an association with the use of task organization mechanisms.

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INTRODUCTION

For many organizations, work has become very global. Team members in these organizations work from multiple locations around the globe often spanning time zones. These global teams need to bridge distance and time boundaries to get the job done (2003; Lu et al., 2003; Orlikowski, 2002; Watson-Manheim et al., 2002a). This presents substantial challenges to technical teams working on projects with highly interdependent task activities that require coordination (Herbsleb & Grinter, 1999; Van de Ven et al., 1976). Distance and time separation are important when studying dispersed teams mediated by technology (Bullen & Bennett, 1993), but while there is a growing body of research on global, geographically dispersed and virtual teams (Ahuja et al., 2003; Armstrong & Cole, 2002; Carmel, 1999; Cramton, 2001; DeSanctis & Monge, 1999; Griffith et al., 2003; Herbsleb & Mockus, 2003; Hinds & Bailey, 2003; Jarvenpaa & Leidner, 1999; Kiesler & Cummings, 2002; Majchrzak et al., 2000; McDonough et al., 2001; Mortensen & Hinds, 2002; Olson & Olson, 2000; Powell et al., 2004; Van den Bulte & Moenaert, 1998; Watson-Manheim et al., 2002b), the study of time separation has gone largely neglected. We argue in this paper that time separation is a great challenge to geographically dispersed teams, particularly as more time zones and larger time zone differences are represented in a team.

When team members are separated by distance, their interactions are largely mediated by tool-mediated communication and information technologies (DeSanctis & Poole, 1997; Drucker, 1993), which reduces the benefits of co-presence (e.g., frequent and spontaneous interaction, availability of contextual reference, presence awareness, workspace awareness, etc.). However, as business environments have become more global, individuals and organizations have become

more adept at bridging geographic distance with communication and other collaboration tools. It is not uncommon to see meetings of geographically dispersed members facilitated by tools like videoconference, white boarding, and shared data repositories. Furthermore, collaboration tools designed to provide benefits of co-presence like presence awareness (Boyer et al., 1998; Handel & Herbsleb, 2002), task awareness (Greenberg et al., 1996) and expertise browsing (Mockus & Herbsleb, 2002) have made it easier for team members to locate each other and retrieve task status information when needed. As these tools are adopted and used more widely in organizations, it is important to understand their consequences in global teams.

While scholars and managers both emphasize best practices learned from experience and effective use of collaboration and communication tools, it is critical not to overlook the time boundaries that team members face. In particular, time separation can have a strong impact on the project workflow. When teams span two or three time zones, its members can resort to tactics like shifting work hours, flextime, and use of mobile communication devices. However, as more time zones are represented in a team, the challenge of coordinating workflows becomes more substantial (Espinosa & Pickering, 2006). We argue that time differences are an important contributor to coordination processes and outcomes, which in turn affect the success of a team.

In this study, we address the following research question: *how is time separation associated with coordination and project outcomes in global teams?* In the next section of this paper we develop the theoretical foundation and formulate our research framework and hypotheses. Then we discuss our study methodology and present our results. Finally, we discuss the implications of our findings followed by our concluding remarks.

THEORETICAL FOUNDATIONS

Research Framework

Our research framework follows the typical “input-process-outcome” (I-P-O) model widely used to study effectiveness in groups that interact with technology (McGrath & Hollingshead, 1994) and for virtual teams (DeLone et al., 2005; Powell et al., 2004). While we speculate that other more complex mediation and moderation frameworks may be appropriate to study the effect of time separation, we prefer to use the simpler I-P-O framework given the lack of theoretical development in this area of research. We illustrate our framework in Figure 1 and provide the corresponding theoretical foundations below. In essence, our proposed framework models geographic configuration as an exogenous input representing the context in which the global team operates. We include team size and task dependencies as control variables because the choice of coordination mechanisms will be affected by the number of members and type of dependencies that exist among task activities (March & Simon, 1958; Thompson, 1967; Van de Ven et al., 1976). We then include coordination processes, such as communication, task organization, and team knowledge. Finally, we include two types of outcomes, an intermediate coordination outcome, and subsequent project outcomes. We next develop the theoretical foundations and hypotheses, following this framework in Figure 1 from left to right.

Geographic Configurations

Global teams are often separated by various team boundaries (e.g., time, distance, cultural, organizational, etc.) that make it more difficult for members to work together (Espinosa et al., 2003; Orlikowski, 2002; Watson-Manheim et al., 2002a). In this study we focus on two of these boundaries, which are of importance to geographically dispersed teams whose interaction is mediated by technologies: time and distance (Bullen et al., 1993; McGrath et al., 1994). Our main goal is to develop a better understanding of the effect of time separation. Researchers of global, virtual, and geographically dispersed teams have studied the effect of geographic dispersion, but most studies have treated dispersion as a dichotomous variable – co-located or

distributed. More recent studies have attempted to better define the concept of “virtualness” (Griffith et al., 2003; Kirkman & Mathieu, 2005; Lu et al., 2003) and geographic dispersion (O’Leary & Cummings, 2002). Griffith et al. (Griffith et al., 2003) proposed “percentage of time apart in the task” as one of three fundamental dimensions of “virtualness”. However, the concept of “time apart” is difficult to define because a large portion of team work is done through electronic media and while the team may be apart, it may be connected and interacting synchronously. Kirkman and Mathieu (Kirkman et al., 2005) took this concept one step further and proposed “synchronicity” as one of their three fundamental dimensions of “virtuality”, which more appropriately distinguishes between “real-time” (i.e., synchronous) and “lagged-time” (asynchronous) interaction.

Measuring virtuality or virtualness along time dimensions like “time apart” and “synchrony” are a step in the right direction, but these concepts are difficult to apply in larger teams in which some members may be co-located and others apart, and in which some members may interact predominantly in real-time while others may interact in lagged-time. This difficulty gets compounded when time separation (e.g., time zones, different holiday and weekend schedules, different work shifts) is considered. Time separation brings about additional challenges that are distinct from pure distance separation, such as: (a) it is not clear what is the best way to measure distance (e.g., geographic, travel time, number of sites represented, dichotomous), whereas time separation is mathematical and can be measured with a work time overlap index (O’Leary et al., 2002) or some other time distance measure based on time zones and work hours; (b) the number of sites represented on a project does not severely affect the team interaction synchrony in contexts where members are in the same time zone but not geographically separated, whereas the number of time zones creates severe workflow coordination challenges, so the timing of the interaction is critical (Espinosa et al., 2006); and (c)

teams have a choice of synchronous and asynchronous communication options in contexts where work hours are similar, whereas often the choice for synchronous communication in time-separated contexts requires waiting for an overlap in work hours.

Perhaps a good way to illustrate how various combinations of distance and time separation affect the dynamics of a team is to look at the various geographic configurations often encountered in global teams. Prior research has suggested that the geographic configuration of a global team can be characterized by its North-South (without time zone differences) dispersion and East-West (with time zone differences), the number of sites, and by its dispersion balance (i.e., difference in size across sites) (O'Leary et al., 2002). An interview study with 23 technical team members at a large semiconductor company found some of these configurations in global teams and that these can be classified into five major categories: Co-located (CO) – a benchmark; North-South (NS) – more than one location represented in a team with no or little time zone differences; East-West (EW) – two or three locations with substantial time differences; Dispersed-Balanced (DB) – more than three locations with substantial time differences and members evenly spread across sites; and Dispersed-Unbalanced (DU) – similar to distributed-balanced, but with a high concentration of members in one site and one or very few members in each of the other locations (Espinosa et al., 2006).

Classifying teams into these configurations with precision is difficult because, except for CO vs. NS teams, distance and time separation are often highly correlated. Even with NS teams, many of their members are co-located and the distribution across locations could be balanced or unbalanced. Finally, the classification of a team as balanced or unbalanced requires the application of subjective criteria. Consequently, it is more useful to characterize these geographic configurations using continuous variables, such as: distance away, number of time zones and maximum time zone span represented in a team, and time zone balance (evenly distributed vs.

unbalanced). These variables can later be used to categorize teams into specific configuration types for analysis and discussion. For example, an EW configuration will be characterized by a small number of time zones, large time zone span, and large distance away. In contrast, a DU configuration will be characterized by a large number of time zones, a large time zone span and distance away, and a low time zone balance (i.e., unbalanced).

These distinctions are important. For example, teams in EW configurations learn to cope with the exact time differences across sites and resort to tactics (e.g., flex time, work at home, arrive/leave early/late, etc.) to increase the work time overlap and interaction synchrony among members. However, coordination challenges increase rapidly as more time zones are represented within a team because the window for synchronous interaction is severely reduced and the timing of interaction becomes more difficult to plan. For example, a prior theoretical model predicts that differences in time zones, coupled with the tool-mediated communication and more asynchronous interaction, create vulnerabilities in the team leading to miscommunication and need for re-work, resulting in coordination delays (Espinosa & Carmel, 2004), which is consistent with coordination theory (Malone, 1987). However, for a given number of time zones represented in a team, we posit that neither the time zone span or the distance separation within the team will not be associated with increased delays because team members learn how to adjust to these differences. Thus, we don't expect substantial coordination delays in DU, CO or NS configurations because when the majority of the team is in one location or time zone, the timing of workflows can be more easily adjusted to manage dependencies of the core site with the few members in satellite locations. Thus, we posit:

Hypothesis 1: The number of time zones represented in a team is positively associated with increased communication delays.

Hypothesis 2: Controlling for number of time zones, the maximum time zone spanned by a team is not associated with increased communication delays.

Hypothesis 3: Controlling for number of time zones and maximum time zone span, distance away is not associated with increased communication delays.

Hypothesis 4: Teams whose members are more evenly distributed across time zones (i.e., balanced) are associated with more communication delays than teams whose members are more unevenly distributed across time zones.

Coordination Processes

Coordination theory defines coordination as the management of dependencies among activities in the task (Malone & Crowston, 1994). Basically, if two people are working together and their individual task activities are interdependent, the respective dependencies need to be effectively managed. Consistent with this, we define coordination process as the actions taken to manage dependencies among task activities. These include the use of coordination mechanisms like procedures or tools used to manage task dependencies. The classic organizational research literature suggest that teams use two main kinds of mechanisms to coordinate their work: communication and task organization (March et al., 1958; Thompson, 1967; Van de Ven et al., 1976). Communication is most effective when managing less routine task dependencies, though communication is also necessary when a routines change (e.g., system breakdowns, schedule delays). Task organization mechanisms include things like division of labor, plans, specifications, and schedules, which are effective to manage the most routine task dependencies.

We argue that because time separation affects the ability for team members to interact synchronously, configurations involving more time separation will be associated with lower levels of face-to-face communication and higher levels of tool-mediated communication and the use of task organization mechanisms to coordinate their work. Thus, we posit:

Hypothesis 5: Teams operating in more time zones have lower use of face-to-face communication to coordinate their work.

Hypothesis 6: Teams operating in more time zones have higher use of tool-mediated communication to coordinate their work.

Hypothesis 7: Teams operating in more time zones have higher use of task organization mechanisms to coordinate their work.

The team cognition research literature also suggests that members utilize various forms of team knowledge (e.g., shared knowledge, transactive memory, team awareness) to manage their dependencies more effectively. When members share knowledge about the team or the task, this knowledge helps them develop accurate expectations about future states of the task and the possible actions of other team members (Cannon-Bowers, Salas et al. 1993; Klimoski and Mohammed 1994), and share information more effectively (Krauss and Fussell 1990; Cramton 2001), thus making their actions more coordinated. Similarly, more fleeting knowledge like team awareness helps members coordinate because this awareness enables members to perceive and comprehend what is happening in their shared task environment (Wellens 1993; Endsley 1995), providing them with up-to-the-minute contextual information relevant to their individual activities (Dourish & Bly, 1992). This type of coordination has been referred to as “implicit coordination” or the “synchronization of member actions based on unspoken assumptions about what others in the group are likely to do” (Wittenbaum & Stasser 1996), thus making it important under conditions where communication is costly (e.g., in fast-paced real-time tasks, across time and distance, etc.). While team cognition can be most helpful in conditions in which communication is impaired (e.g., time and distance separation), it is precisely these conditions that make it difficult for teams to develop team knowledge about team members and the task.

Hypothesis 8: Teams operating across larger distances have lower levels of team knowledge.

Hypothesis 9: Teams operating in more time zones have lower levels of team knowledge.

Coordination Outcomes

Consistent with coordination theory we define coordination outcomes as the resulting state of coordination in the project. That is, a project will be successfully coordinated if all key dependencies among its task activities have been effectively managed. Different tasks have different kinds of dependencies. For example, well coordinated technical projects will not only exhibit well managed technical dependencies, but temporal and process dependencies as well. Technical coordination is important when various parts of a product are being developed by different team member, which then need to be integrated into a single product. Technical dependencies are effectively managed when there are no integration problems. However, temporal coordination is also important when teams need to meet deadlines. Even when various parts integrate seamlessly, ineffective management of temporal dependencies can throw a project in disarray because the individual parts are not delivered for integration according to schedule. Finally, process coordination is important in complex technical tasks to ensure that the established work processes are adhered to. Ineffective management of process dependencies result in things like priority conflicts, missed steps, roadblocks that require escalation to upper management, and misunderstandings, among other things.

As we discussed before, effective project coordination is accomplished explicitly through communication, task organization (March et al., 1958; Thompson, 1967; Van de Ven et al., 1976) and team knowledge (Cannon-Bowers et al., 1993; Klimoski & Mohammed, 1994). We are not aware of any empirical studies that have evaluated which types of coordination processes are most effective for which contexts. The empirical evidence on the effect of communication is

mixed. While some studies have shown the importance of communication for team coordination (Kraut & Streeter, 1995; Perry et al., 1994) other studies have found no effects, mixed effects or even negative effects (Kiesler et al., 1994; Wholey et al., 1996). We speculate that the effect of communication on coordination is difficult to assess because, while communication is needed to coordinate, uncoordinated teams need to communicate actively to resolve their issues. Because of this mixed evidence, we don't offer specific hypotheses related to the effect of communication on coordination outcomes, but we include communication variables in the model for control and exploratory purposes.

The evidence on the effectiveness of task organization mechanisms and team knowledge is a bit clearer. Empirical evidence has shown positive effects of task organization mechanisms custom tailored to the task (e.g., configuration management systems) (Grinter, 2000). Thus, we posit:

Hypothesis 10: The use of task organization mechanisms is negatively associated with coordination problems.

Finally, empirical research suggests that groups benefit from access to well-developed and accurate team knowledge about the task (Mathieu et al., 2000; Nelson & Coopridge, 1996) and about team members (e.g., transactive memory, access to team member expertise) (Faraj & Sproull, 2000; Lewis, 2003; Liang et al., 1995). We don't offer hypotheses about the effect of team knowledge on coordination outcomes because the positive effects of team knowledge have been demonstrated empirically, and because it is not the specific focus of our study, but we include this variable for control purposes.

Project Outcomes

When task activities can be carried out relatively independently, project outcomes are mostly the result of individual activity. In addition to individual performance, project

coordination is important for tasks that have tightly coupled and complex interdependencies among task activities (March et al., 1958; Thompson, 1967; Van de Ven et al., 1976). Because the link between coordination outcomes and project outcomes is not the focus of this study, we do not offer any hypotheses about this relationship. However, we include this relationship in our research framework to establish a link between geographic configurations, coordination processes, coordination outcomes, and project outcomes. Because the tasks in the present study are mostly technical and complex, we anticipate that coordination outcomes will affect project outcomes.

STUDY METHOD

Study Context and Sample

This study was conducted through a web survey of team members working on technical projects for a semiconductor manufacturing company in 54 global locations in 23 countries country (Belgium, Brazil, Canada, China, Costa Rica, Denmark, France, Germany, India, Ireland, Israel, Japan, Malaysia, Mexico, The Philippines, Poland, Russia, Singapore, South Korea, Taiwan, The United Kingdom and the United States). The target population for our survey included primarily technical engineers and software engineers, but also included project managers, project coordinators, business analysts, and systems analysts, among others, from multiple technical and business departments. We sent a description of our research to a random sample of approximately 4000 project managers across the company. Of the 380 project managers who expressed interested in participating, approximately 300 of them provided a description of their project and the names of their team members. We used this information to assemble a list of 2318 participants to whom we emailed requests to participate and 1,311 of them completed surveys, providing a response rate of 56.6%. The 1,039 respondents who where project managers or project members are the main focus of our study and were thus categorized

as “core” members. The remaining 272 were project advisors, outside experts and other stakeholders with limited involvement in the project, who were categorized as “non-core” members.

The web survey instrument was dynamically generated with scripts to include items about the projects they worked on (i.e., project level items) and about the specific team members who worked on that project (i.e., dyad level items). These survey items are displayed in Tables 1, 2, and 3.

Study Variables

The descriptive statistics and correlation matrix for all our study variables are presented in Table 5. We now describe each variable.

Geographic Configurations. We created several variables to capture all key descriptors of geographic configuration that could be used to classify teams into the 5 configurations we described earlier: Co-Located (CO), North-South (NS), East-West (EW), Distributed-Balanced (DB), and Distributed-Unbalanced (DU). The CO and NS configurations can be captured with a distance away variable measured with a Likert scale ranging from 1 (same room) to 5 (different countries), which has been used in prior studies of global teams (Cummings, 2004). We also included two important measures of geographic dispersion: the number of locations and the number of time zones represented in a team (O'Leary et al., 2002). To measure the amplitude of this dispersion we included two other measures: the average time zone difference across all dyads in the team (O'Leary et al., 2002) and the maximum time zone spanned in the team. Finally, we included a measure of dispersion balance to capture the extent to which team members are widely distributed across locations or concentrated in one or few locations (O'Leary & Mortensen, 2005). We computed this variable by dividing the average number of members in

each time zone by the largest number of members in one time zone, such that a measure of one indicates perfect balance and lower numbers represent imbalance.

One difficulty of including these variables in analysis models is the high correlation often found among them. For example, as table 5, the correlation between number of locations and number of time zones is 0.86. Similarly, the correlation between the maximum time span and average time difference is 0.58. Therefore, we used factor analysis with Variamax rotation to better understand the commonalities and underlying structure of these variables. We also included team size in this analysis because of its high correlation with number of locations and time zones, along with the average number of members per time zone. We found an underlying structure for these variables consisting of two variable using a cutoff criterion of eigenvalue less than one. Since the next factor had an eigenvalue close to 1 we forced the solution to three factors: time zone dispersion (number of locations and number of time zones), time zone magnitude (maximum time zone span and average time difference among sites) and team size (team size—core members only and number of members per time zone). Distance away and time zone balances emerged as individual variables with factor loadings of 0.65 or less. Therefore, we removed these two variables and re-run factor analysis and computed the respective factor scores. The results are shown in Table 4.

Because we are more interested in understanding the effect of individual configuration variables, we decided to select the most meaningful variables for our study from each factor, rather than grouping variables into factors. Therefore, we selected number of time zones, maximum time zone span, and team size for our analysis, and included the two variables without factor loadings, distance away and time zone balance. However, to better understand the effects of geographic configuration we conducted post-hoc analysis using some of the alternative variables for each factor as well as the factor scores. We included core team size as a control

variable because research has shown that it can have a negative effect on coordination (Herbsleb et al., 2003). As Brooks (1995) pointed out a team of size n has $n(n-1)/2$ possible communication and interdependency links, so the coordination overhead increases substantially with team size. Finally, we also included project length (in months) as a control variable.

Dyadic Variables. For *Task Dependencies*, we used three dyadic survey items (“tasks this contributor performed were related to tasks you performed”, “this contributor depended on you for information or materials needed to complete their work”, and “you could not accomplish your tasks without information or materials from this contributor”) (Cronbach- $\alpha=0.94$). For *Communication*, we used the five dyadic communication items shown in Table 2, which were reduced through factor analysis into two variables explaining 79.0% of the variance, *Face-to-Face Communication* (Cronbach- $\alpha=0.87$) and *Tool-Mediated Communication* (Cronbach- $\alpha=0.90$). We were hoping to discriminate between synchronous and asynchronous communication tools, but all tool-mediated items loaded together. For *Coordination Delay* and *Team Knowledge*, we used the nine dyadic items shown in Table 3, which were reduced through factor analysis into 2 variables explaining 77.0% of the variance. The items for the first variable, *Coordination Delay* (Cronbach- $\alpha=0.83$), loaded as expected. The second variable was *Team Knowledge* (Cronbach- $\alpha=0.94$). We were expecting to discriminate between situational and more permanent team knowledge, but all items loaded together, so we combined them into a single variable.

Project-Level Variables. For *Task Organization* (Cronbach- $\alpha=0.71$) and *Coordination Problems* (Cronbach- $\alpha=0.90$) we used twelve project-level items shown in Table 1, which were reduced through factor analysis into 2 variables explaining 58.9% of the variance. Items in the task organization variable loaded as expected. For coordination problems, we were expecting to be able to discriminate between technical, temporal and process coordination problems, but these

items loaded together in a single factor, so we combined all items into one variable. For project outcome variables, we used nine project-level items, also shown in Table 1, which were reduced through factor analysis into 2 variables explaining 64.9% of the variance. The items in these two variables, *Project Impact* (Cronbach- α =0.87) (e.g., realize the maximum impact of the achievement, identify where else achievement could have impact) and *Project Performance* (Cronbach- α =0.81) (e.g., within budget, within schedule, and met requirements) loaded as expected.

Data Analysis Method

The survey items were first analyzed with confirmatory factor analysis with Varimax rotation of individual responses. Dyadic and project-level items were analyzed separately. Only items with similar scales were analyzed together in each factor analysis model. All items loaded in the respective factor variables as expected, except as noted above. The items were then averaged to the project level. For project level items, only responses from core members were aggregated. For dyadic levels, only responses for core-core dyads were aggregated. After aggregating the data, we discarded all projects that were represented by less than 50% of the core team members, which yielded 71 teams with an average size of 8 members.

Because *Project Impact* and *Project Performance* were modeled as dependent variables, and *Task Organization* and *Coordination Problems* were modeled as independent variables, we further evaluated these variables for problems with common method variance. We first re-ran factor analysis of the individual responses, but this time we used all project-level items. As Table 1 shows, the factors discriminated well across variables, yielding similar loadings, thus providing some assurance that problems with common method variance were reduced. We took further steps to minimize problems of common method variance. First, most process variables were collected at the dyad level, whereas all other variables were collected at the project level, thus

providing different levels of granularity in the data. Second, all coordination process and coordination outcome variables were averaged for core team members only, whereas the two project outcome variables were computed using responses from both core and non-core members. While non-core members are usually not as familiar with the coordination processes used by core members, they are often in a stakeholder position that enables them to judge project outcomes.

We analyzed the data using ordinary least squares (OLS) regression analysis. We constructed several models following the framework illustrated in Figure 1. The regression models were formulated in a structural fashion first by regressing all geographic configuration variables on project length. We then regressed all geographic configuration variables and project length on the coordination process variables, but controlling for team size (Brooks, 1995) and task dependencies (Van de Ven et al., 1976) because these are two variables known to affect coordination. We then regressed all coordination process variables on coordination outcomes, and then regressed coordination outcomes on project outcomes. Each subsequent regression model included all variables from the prior models, so that all direct and mediation effects could be evaluated (Baron & Kenny, 1986). We inspected all variables for multicollinearity using variance inflation factors (VIFs) and all variables had VIFs well below the recommended limit of 10 (Marquardt, 1970). However, when we inspected the models for multicollinearity using the condition index, three of the latter models had conditions neighboring the recommended limit of 30 (Belsley et al., 1980). Condition indices below 100 are not considered severe, particularly when VIFs are low because none of the standard errors are inflated, so the coefficient estimates are stable. Nevertheless, we further reduced the condition indices in these three models by removing all variables that were not significant at the $p < 0.20$ level. We chose this level to be conservative and retain some of the original variables that were not significant. We conducted

Wald-F tests to compared all the reduced models and their respective full models and none of the full models added significant predictive power over the reduced models. We also tested all the group level variables using intra-class correlation (ICC) statistics and all variables exhibited group effects at the $p < 0.001$ level, which is better than the recommended $p < 0.25$ level needed to justify aggregation of individual variables to the group level (Myers, 1972).

RESULTS

Our results are presented in Table 6 and illustrated in Figure 2. In support to hypothesis 1, our results show that teams operating in more time zones experience longer communication delays ($\beta = 0.12$, $p = 0.020$). Also, in support of hypothesis 2, we found no significant association between the maximum time zone span and communication delays ($\beta = 0.01$, $p = 0.566$). Similarly, in support of hypothesis 3 we found no significant association between distance away and communication delay ($\beta = 0.07$, $p = 0.170$). However, we found no support for hypothesis 4 in that balance in geographic distribution balance had no effects on communication delay.

Hypothesis 5, that face-to-face communication is lower in teams operating in more time-separated was not supported and only distance away was negatively associated with face-to-face communication ($\beta = -0.35$, $p < 0.001$). We did not find support for hypothesis 6 in that teams operating in more time zones were not associated with more tool-mediated communication. Only team size ($\beta = -0.03$, $p = 0.026$) and task dependency ($\beta = 0.31$, $p = 0.005$) were associated with tool task dependency. Also, we did not find support for hypothesis 7 in that the number of time zones was not associated with the use of task organization mechanisms. Surprisingly, no geographic configuration variable was associated with the use of task organization. Similarly, we did not find support for hypotheses 8 or 9 in that we found neither distance away or number of time zones was associated with team knowledge. Distance away was negatively related to team knowledge, but the coefficient's significance level was borderline ($\beta = -0.08$, $p = 0.100$). Time zone

balance was also moderately associated with team knowledge ($\beta=0.49$, $p=0.096$) suggesting that more even distribution of members across time zones promotes more team knowledge development than when members are concentrated in one or a few time zones. Also, only task dependency was positively associated with team knowledge ($\beta=0.43$, $p<0.001$), suggesting that interdependent tasks help teams develop shared knowledge.

Consistent with hypothesis 10, we found that the use of task organization mechanisms was negatively associated with coordination problems ($\beta=-0.74$, $p<0.001$). In addition, we found that communication delay was positively associated with coordination problems ($\beta=0.33$, $p=0.004$). However, the maximum time zone spanned by a team was positively associated with coordination problems ($\beta=0.03$, $p=0.031$). This result is very interesting in that both the number of time zones and larger time zone differences create coordination problems, but the effect of time zone span is direct, whereas the effect of number of time zones is indirect and mediated by communication delay. Also interestingly, tool-mediated communication was positively associated with coordination problems ($\beta=0.27$, $p=0.004$). We suspect that causality may be in the opposite direction in that teams with uncoordinated projects resort to more tool-mediated communication to resolve their issues. Finally, task dependency was positively associated with coordination problems ($\beta=0.35$, $p<0.001$). One interesting observation about this result is that task dependency is beneficial for team knowledge development but it creates more coordination problems because more dependencies need to be effectively managed.

Coordination problems had a strong negative association with project performance ($\beta=-0.64$, $p<0.001$) and project impact ($\beta=-0.20$, $p<0.016$). This is an important finding because it is not only consistent with coordination theory in that interdependent tasks require coordination, which provides validity to our findings, but it also helps us establish a path between geographic configurations and project outcomes. Task dependency also was positively associated with

project performance ($\beta=0.34$, $p<0.001$), suggesting that the increased level of team interaction generated by the need to manage dependencies pays off when it comes to completing the project within budget, within schedule, and by meeting requirements. Once again, task dependency appears to create coordination problems, but it has direct beneficial effects on team performance, team knowledge development and reduced communication delay ($\beta=-0.27$, $p=0.006$). However, task dependency was not associated with project impact but team knowledge was marginally associated with project impact ($\beta=0.18$, $p=0.059$), suggesting a weak indirect association between task dependency and project impact mediated by team knowledge. Interestingly, the number of time zones was positively associated with project impact ($\beta=0.07$, $p=0.037$) but the maximum time zone span had a negative association ($\beta=-0.04$, $p=0.002$). This result suggests that having team members in multiple time zones is beneficial for project impact, but that for a given number of time zones represented, the magnitude of the time separation offsets these benefits.

DISCUSSION

The principal findings of this study are twofold. First, as one would expect, face-to-face communication decreases as teams become more geographically dispersed. However, this reduced communication is due to pure geographic distance and there are no further decreases in face-to-face communication as the team spans more time zones. On the other hand, pure geographic distance had no effect on communication delay, which was only affected by the number of time zones and task dependency. Therefore, tasks that require substantial face-to-face communication have more to benefit from co-location, but that for a given geographic distance configuration, further time zone separation has no incremental effect. On the other hand, tasks that can be affected by tightly coupled dependencies suffer substantial communication delay and coordination problems as time separation increases.

Especially noteworthy is the role of task dependencies in coordination of geographically dispersed teams. Task dependency had a positive effect on tool-mediated communication, team knowledge, and project performance, and it reduced communication delay. But it had detrimental effects on coordination outcomes (i.e., positively related to coordination problems). This has important implications for task design, geographic dispersion decisions and collaboration tool design. Some aspects of task dependency are inherent in the task, whereas other aspects can be the result of de-coupling decisions made to reduce task dependencies among team members. While this de-coupling has beneficial effects on coordination it has a negative effect on the use of asynchronous collaboration tools, it prevents team knowledge development and has a direct detrimental effect on team performance. These results underscore the importance of task design and healthy levels of task interdependencies among team members, which naturally need to be well managed. It is important to note that we use the term “healthy levels” when we discuss task dependencies because we believe that being excessively dependent on other team members can be detrimental. While moderate levels of task dependency among members can foster team communication and team knowledge development, very high levels of task dependency can create roadblocks when the work of one team member cannot continue until the work of another member is finished. This possibly curvilinear effect of task dependencies needs to be further investigated.

Also surprisingly was the fact that none of the geographic configuration or task dependency variables were associated with the use of task organization mechanisms, especially given the fact that task organization is the most significant predictor of reduced coordination problems. However, this is consistent with recent findings of a qualitative study (Espinosa et al., 2006) that found that global team members tend to select a mix of coordination mechanisms based on personal preference and what they think will work for the task. And once a team adopts

certain coordination mechanisms in a project, they tend to use them consistently with most team members, regardless of location, which may explain the lack of variance in the use of task organization mechanisms across configurations. This has strong implications for global team managers who need to implement effective policies and practices that foster the implementation and use of effective task organization mechanisms, which is particularly important when global boundaries make it more difficult for teams to communicate to coordinate their work. Best practices aimed at developing, implementing and using effective task organization mechanisms can have large payoffs in the form of reduced coordination problems. While distance separation reduces the choice of communication options available to the team, this reduction is not as substantial as the reduction caused by time separation. On the other hand, the mix of communications options, especially synchronous, is severely reduced when the team members are separated across multiple time zones. It is precisely under these conditions that task organization can be most effective because it helps teams coordinate with less communication. In fact, the use of task organization processes custom tailored to the specific needs of a task (e.g., configuration management) could be quite effective at helping teams coordinate in large scale tasks. Another important way to achieve higher levels of project coordination is to minimize coordination delays, which can be accomplished by having healthy levels of task dependencies among members, particularly those in distributed balanced configurations.

As we predicted, teams with more time-separated configurations experienced more coordination delays than co-located teams, which supports intuition. Not as intuitive, and perhaps more interesting is the fact that teams in North-South configurations did not experience more coordination delays than co-located teams. These teams may have learned how to coordinate their work in ways that it didn't create significant delays. For example, geographic distance may have been an incentive for teams to establish effective collaboration policies and

adhere to them. Theoretically, time separation could actually reduce delays if the workflows were properly synchronized because one site could do the work while the other site ‘sleeps.’ However, when these workflows cannot be tightly synchronized, delays should be expected. In contrast, members of co-located teams may feel that such policies are necessary because team members can always walk over to their colleagues’ offices to coordinate things when needed, but also causing possible interruptions and delays.

Finally, both project outcome variables, project performance and project impact, were negatively associated with coordination problems. This finding, while intuitive and consistent with coordination theory (Malone et al., 1994), establishes an important link between our other findings and ultimate project outcomes. Because the tasks we studied are generally technical, complex and highly interdependent, coordination is important. Task dependencies also had a positive association with project performance.

Finally, because we originally had several correlated variables to measure various aspects of geographic dispersion, we selected team size in favor of number of team members per time zone and maximum time zone span in favor of average time zone difference. To ensure that we selected the most significant predictor variables we re-ran all our regression models by replacing our chosen variables with members per time zone and average time zone difference. Most results were unchanged and these alternative variables were mostly non-significant, except that the number of time zones ($\beta=-0.156$, $p=0.009$) and the number of members per time zone ($\beta=-0.082$, $p=0.023$) became significantly associated with tool-mediated communication, suggesting that more time zones with more members in those time zones experience reduced use of tool mediated communication media. Similarly, we re-ran the regression models by replacing our original choice of geographic dispersion variable with the orthogonal factor scores obtained when we factor analyzed these variables. Again, most results were unchanged, except that time

zone magnitude ($\beta=0.133$, $p=0.031$) and time zone balance ($\beta=0.582$, $p=0.048$) were positively associated with communication delay, and the number of time zones was negatively associated with tool-mediated communication ($\beta=-0.185$, $p=0.016$). These results confirm the robustness of our findings with our original selection of variables, but it also provides further evidence of the negative association of time separation with communication delay and tool-mediated communication.

Limitations and Conclusions

Our study has several limitations. First, while our participant sample is quite large and from various technical domains, all employees come from a single company. Naturally, our findings are limited to the idiosyncrasies of this company. For example, other organizations without substantial global collaboration experience may encounter more severe coordination problems across distance and time. Second, the study was conducted using survey methods, so it is subject to problems of common method variance. Nevertheless, we minimized this problem by collecting some variables at the dyad level and some at the project level, and by including responses from non-core members in the two project outcome variables, while only including responses from core members in all other variables. Third, our measures of geographic configuration are based on the actual geographic location of each member. In practice, team members resort to tactics like shifting their work hours, working from home, and making themselves more available by using mobile communication devices to extend the work time overlap across sites, thus increasing the window for synchronous interaction. Therefore, the real time separation between members is generally smaller than the actual time separation based on geography. Further research is needed to better understand the effect of these tactics.

Despite these limitations, our study makes important contributions to theory and practice. Our theoretical contribution is a foundation for understanding the effects of time separation in

global collaboration. This foundation is centered on issues of geographic configurations, coordination processes, coordination outcomes, and project outcomes to further study the impact of time separation in global teams. To our knowledge, this is the first empirical study to investigate the effect of time separation using quantitative methods. More importantly, our study is the first to study global teams using various predictor variables measuring different aspects of geographic dispersion (e.g., distance away, time zone balance, time zone span, number of time zones) in a comprehensive way, thus providing a more effective way to tease out the effects of distance from time separation, which are often confounded in studies. Finally, our study makes an important contribution to the literature on geographically dispersed, virtual, and global teams, in which much has been written about the effects of distance separation, without examining more closely the often confounded effect of time separation.

Our study also has important implications for practice for both managers and collaboration tool designers. For managers, our study suggests that effective collaboration across time zones requires paying close attention to task dependencies and the sources of coordination delays. Well designed tasks that contain healthy levels of task dependencies and foster communication help reduce coordination delays. Our study also suggests that managers need to pay attention to how and which task organization mechanisms are adopted. Geographic configuration and task dependencies don't appear to influence team members' decisions to use task organization mechanisms proactively to coordinate their work across time zones and prior research suggests that these choices are based more on personal preferences. However, our study strongly suggests that task organization is the most effective way to reduce coordinate problems in global contexts.

For collaboration tool designers, our study suggests that these tools need to incorporate features that will enable teams to organize their workflow and manage their tasks. These tools

need to be configurable so that the most appropriate task organization mechanisms are evident to team members to foster their use. It may be beneficial to introduce explicit task organization mechanisms custom tailored to the task especially when more time zones are represented in a team. The use of workflow automation to track status and make workflows more explicit may reduce the amount of effort to manually coordinate the work. It can also reduce the number of real-time meetings needed to coordinate work and maximize the time overlaps because people come to the meeting prepared to discuss key issues and topic areas, rather than using the time to catch up on status, which can reduce coordination delays.

It is also important for collaboration tools to include features to help cope with the coordination delays brought about by time separation. Collaboration tool designers are encouraged to explore tool features that help team members select an optimal level of task dependencies that will foster communication and team knowledge, without over-taxing the team with too much coordination overhead. Explicit representation and tracking of task dependencies may foster communication and knowledge sharing. Also, an integrated environment that supports multiple modes of collaboration including real time text, voice, and video, asynchronous coordination, persistent group memory and context, and increased awareness of remote team members, to allow the team to dynamically adjust coordination mechanisms throughout the work day.

Finally, our study supports the need for smart presence and shared context and history about the task and the team. For example, a persistent project and team workspace for documenting project facts and other content, with simultaneous visual representation of each team member with quick links to their bio, resume, publications, contact information, hobbies, interests, country/culture, and current time of day in their location can help reduce foster team knowledge and reduce coordination delays.

In conclusion, in this study we have investigated how various geographic configurations affect coordination and other global project outcomes in teams. Our findings underscore the importance of developing a deep understanding of the effects of time dispersion in teams, which has been largely ignored in the global teams literature. One important conclusion of this study is that time separation and geographic distance both matter in global collaboration.

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Table 1: Factor Analysis of Project Level Survey Items

Team Level Survey Item	Dependent Variables Independent Variables				All Project Level Variables			
	Project	Project	Coord	Task	Project	Project	Coord	Task
	Impact	Perf	Problems	Organiz	Impact	Perf	Problems	Organiz
Cronbach α Reliability	0.87	0.81	0.90	0.71				
We completed the work on schedule/on-time	0.095	0.860			0.091	0.809	-0.274	0.116
We completed the work well within budget	0.132	0.844			0.153	0.821	-0.175	0.053
The final product met requirements	0.248	0.790			0.245	0.756	-0.171	0.143
Extract potential value from the achievement.	0.735	0.097			0.734	0.085	-0.028	0.092
Realize the maximum impact of the achievement.	0.797	0.233			0.784	0.179	-0.167	0.126
Identify where else achievement could have impact.	0.813	0.104			0.800	0.063	-0.092	0.132
Validate achievement by managers outside of project.	0.755	0.110			0.752	0.078	-0.124	0.074
Ensure achievement was aligned with business goals.	0.750	0.276			0.709	0.190	-0.187	0.256
Take advantage of other related achievements.	0.732	0.104			0.725	0.048	-0.129	0.088
We distributed responsibilities so that we could work somewhat independently from each other			-0.144	0.745	0.109	0.085	-0.163	0.746
We established ground rules, routines and meeting schedules to facilitate our team			-0.118	0.812	0.203	0.100	-0.134	0.773
We organized our team in a way that would make us work more effectively across geographies			-0.167	0.786	0.239	0.095	-0.172	0.731
We often had to re-plan things because of missed delivery dates			0.698	-0.117	-0.099	-0.244	0.661	-0.052
We had many problems due to confusion and misunderstanding (by me or by others) about our task requirements			0.751	-0.258	-0.157	-0.055	0.759	-0.203
We had many problems due to non-adherence (by me or by others) with project processes and procedures			0.735	-0.210	-0.128	-0.050	0.745	-0.165
We had many problems due to priority conflicts			0.729	-0.144	-0.110	-0.114	0.719	-0.094
We had several problems that had to be escalated to higher levels in the organization			0.711	-0.074	-0.040	-0.150	0.698	-0.054
We had many instances of substantial roadblocks			0.775	-0.016	-0.039	-0.142	0.758	0.014
We had a lot of redundant work			0.705	-0.209	-0.118	-0.047	0.711	-0.173
We had a lot of integration problems			0.769	-0.149	-0.107	-0.110	0.762	-0.092
We had to do a lot of re-work			0.784	-0.151	-0.155	-0.118	0.770	-0.085

Table 2: Factor Analysis of Dyad Communication Survey Items

Dyad Level Communication Survey Item	Tool	F2F
	Comm	Comm
Cronbach α Reliability	0.90	0.89
<i>I communicated with this team member:</i>		
Face-to-Face Communication	0.216	0.864
Voice Communication (e.g. telephone or voice conference)	0.834	0.262
Text Communication (e.g. instant messenger)	0.678	0.369
Asynchronous Text Communication (e.g. e-mail)	0.800	0.379
Shared Desktop Based Communication (e.g. NetMeeting or WebX)	0.877	0.055
About task-related knowledge	0.738	0.480
To socialize or discuss topics not related to work	0.301	0.854
Through informal or unplanned encounters	0.257	0.856

Table 3: Factor Analysis of Delay and Team Knowledge Survey Items

Dyad Level Delay/Knowledge Survey Item	Team	
	Knowl	Delay
Cronbach α Reliability	0.94	0.83
<i>About each team member:</i>		
Typically it took a long time to get a response from this person.	-0.352	0.733
Our communications required frequent clarification.	-0.164	0.880
We often had to rework tasks beyond what I would normally expect.	-0.191	0.875
I always knew when and where to find this contributor.	0.841	-0.245
I was usually aware of the status of this contributor's deliverables.	0.846	-0.240
I always knew how to interact with this contributor.	0.841	-0.274
I had substantial knowledge about this person's work on this project.	0.885	-0.182
I had substantial knowledge of this person's skills and area of expertise.	0.883	-0.165
I had substantial agreement with this person on the project's vision and strategies.	0.798	-0.283

Table 4: Factor Analysis of Geographic Configuration Variables

	TZ	TZ	Team
	Dispersion	Magnitude	Size
Number of Locations	0.94	0.16	0.08
Number of Time Zones	0.88	0.41	-0.06
Max Time Zone Span	0.21	0.88	0.00
Average Time Difference	0.21	0.86	-0.03
Team Size Core	0.50	0.18	0.81
Members per Time Zone	-0.20	-0.15	0.95
% Cumulative Variance Explained	34.03	63.48	89.77

Table 5: Descriptive Statistics and Correlation Matrix

For response core members > 50%	N	Mean	Std Dev	Correlation Matrix																	
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 Number of Locations	115	4.37	3.02		0.51	0.86	0.31	0.38	-0.54	0.51	-0.06	-0.58	-0.47	-0.37	-0.05	0.37	0.25	-0.30	-0.07	-0.07	0.08
					0.000	0.000	0.001	0.000	0.000	0.000	0.536	0.000	0.000	0.000	0.610	0.001	0.009	0.003	0.461	0.439	0.421
2 Distance Away (survey)	83	4.81	1.35	0.51		0.54	0.23	0.58	-0.32	0.05	-0.39	-0.66	-0.17	-0.30	0.03	0.36	0.20	-0.16	0.10	-0.05	0.14
				0.000		0.000	0.033	0.000	0.003	0.624	0.000	0.000	0.123	0.006	0.808	0.002	0.072	0.144	0.390	0.661	0.213
3 Number of Time Zones	115	2.96	1.49	0.86	0.54		0.53	0.54	-0.57	0.46	-0.19	-0.52	-0.39	-0.32	-0.09	0.42	0.25	-0.20	-0.13	-0.13	0.05
				0.000	0.000		0.000	0.000	0.000	0.000	0.041	0.000	0.000	0.002	0.343	0.000	0.010	0.052	0.173	0.179	0.627
4 Max Time Zone Span	115	6.41	3.84	0.31	0.23	0.53		0.58	-0.40	0.26	-0.10	-0.31	-0.20	-0.15	-0.04	0.17	0.10	-0.15	-0.14	-0.22	-0.16
				0.001	0.033	0.000		0.000	0.000	0.005	0.278	0.003	0.044	0.140	0.700	0.132	0.309	0.142	0.130	0.017	0.091
5 Average Time Difference	103	2.8	2.34	0.38	0.58	0.54	0.58		-0.37	0.20	-0.15	-0.40	-0.21	-0.31	-0.07	0.37	0.15	-0.16	-0.16	-0.21	-0.04
				0.000	0.000	0.000	0.000		0.000	0.046	0.118	0.000	0.036	0.002	0.498	0.001	0.145	0.115	0.096	0.032	0.685
6 Time Zone Balance (Avg/Max in TZ)	115	0.8	0.21	-0.54	-0.32	-0.57	-0.40	-0.37		-0.55	-0.13	0.49	0.35	0.46	0.03	-0.19	-0.11	0.26	0.12	0.04	0.10
				0.000	0.003	0.000	0.000	0.000		0.000	0.178	0.000	0.000	0.000	0.730	0.097	0.275	0.011	0.214	0.635	0.284
7 Team Size Core	115	8.45	6.57	0.51	0.05	0.46	0.26	0.20	-0.55		0.68	-0.36	-0.47	-0.45	-0.08	0.21	0.15	-0.48	-0.15	-0.10	-0.10
				0.000	0.624	0.000	0.005	0.046	0.000		0.000	0.000	0.000	0.000	0.423	0.066	0.117	0.000	0.116	0.274	0.307
8 Members per Time Zone	115	3.08	2.3	-0.06	-0.39	-0.19	-0.10	-0.15	-0.13	0.68		0.03	-0.21	-0.20	-0.04	-0.05	-0.08	-0.29	-0.04	0.00	-0.11
				0.536	0.000	0.041	0.278	0.118	0.178	0.000		0.745	0.036	0.053	0.657	0.669	0.379	0.004	0.691	0.966	0.225
9 Comm F2F (Comm Mode 1)	92	2.44	0.91	-0.58	-0.66	-0.52	-0.31	-0.40	0.49	-0.36	0.03		0.54	0.41	0.02	-0.40	-0.09	0.31	-0.06	0.01	-0.03
				0.000	0.000	0.000	0.003	0.000	0.000	0.000	0.745		0.000	0.000	0.830	0.001	0.385	0.004	0.599	0.888	0.770
10 Comm Tools (Comm Mode 0, 2-5)	98	2.81	0.67	-0.47	-0.17	-0.39	-0.20	-0.21	0.35	-0.47	-0.21	0.54		0.45	0.17	-0.35	-0.10	0.45	-0.07	-0.01	0.19
				0.000	0.123	0.000	0.044	0.036	0.000	0.000	0.036	0.000		0.000	0.089	0.002	0.330	0.000	0.517	0.897	0.062
11 Team Knowledge (PrjAware & PrjKnl)	96	3.85	0.59	-0.37	-0.30	-0.32	-0.15	-0.31	0.46	-0.45	-0.20	0.41	0.45		0.18	-0.40	-0.19	0.58	0.06	0.27	0.25
				0.000	0.006	0.002	0.140	0.002	0.000	0.000	0.053	0.000	0.000		0.073	0.000	0.074	0.000	0.553	0.007	0.013
12 Project Task Programming (PrjCoord 7-9)	112	4.1	0.52	-0.05	0.03	-0.09	-0.04	-0.07	0.03	-0.08	-0.04	0.02	0.17	0.18		0.00	-0.07	0.14	0.53	0.39	0.47
				0.610	0.808	0.343	0.700	0.498	0.730	0.423	0.657	0.830	0.089	0.073		0.998	0.475	0.174	0.000	0.000	0.000
13 Delay (Crd Delay 0-2)	78	2.08	0.5	0.37	0.36	0.42	0.17	0.37	-0.19	0.21	-0.05	-0.40	-0.35	-0.40	0.00		0.03	-0.28	-0.07	-0.17	0.01
				0.001	0.002	0.000	0.132	0.001	0.097	0.066	0.669	0.001	0.002	0.000	0.998		0.807	0.015	0.552	0.144	0.947
14 Project Length in Months	110	28.9	28	0.25	0.20	0.25	0.10	0.15	-0.11	0.15	-0.08	-0.09	-0.10	-0.19	-0.07	0.03		-0.06	-0.07	-0.12	-0.11
				0.009	0.072	0.010	0.309	0.145	0.275	0.117	0.379	0.385	0.330	0.074	0.475	0.807		0.596	0.449	0.201	0.244
15 Task Dependency (All Dep)	94	3.1	0.65	-0.30	-0.16	-0.20	-0.15	-0.16	0.26	-0.48	-0.29	0.31	0.45	0.58	0.14	-0.28	-0.06		-0.08	0.20	0.26
				0.003	0.144	0.052	0.142	0.115	0.011	0.000	0.004	0.004	0.000	0.000	0.174	0.015	0.596		0.448	0.050	0.011
16 Project Process Coordination (PrjCoord 13-21)	112	3.8	0.62	-0.07	0.10	-0.13	-0.14	-0.16	0.12	-0.15	-0.04	-0.06	-0.07	0.06	0.53	-0.07	-0.07	-0.08		0.59	0.44
				0.461	0.390	0.173	0.130	0.096	0.214	0.116	0.691	0.599	0.517	0.553	0.000	0.552	0.449	0.448		0.000	0.000
17 Project Outcomes, Core & Non Core (PrjCoord 22-24)	114	3.83	0.58	-0.07	-0.05	-0.13	-0.22	-0.21	0.04	-0.10	0.00	0.01	-0.01	0.27	0.39	-0.17	-0.12	0.20	0.59		0.50
				0.439	0.661	0.179	0.017	0.032	0.635	0.274	0.966	0.888	0.897	0.007	0.000	0.144	0.201	0.050	0.000		0.000
18 Project Value, Core & Non Core (ProjChar 19-25)	114	3.64	0.52	0.08	0.14	0.05	-0.16	-0.04	0.10	-0.10	-0.11	-0.03	0.19	0.25	0.47	0.01	-0.11	0.26	0.44	0.50	
				0.421	0.213	0.627	0.091	0.685	0.284	0.307	0.225	0.770	0.062	0.013	0.000	0.947	0.244	0.011	0.000	0.000	

Table 6: Regression Analysis Results

	Communic		Communic		Task		Team		Communic		Coordination		Project		Project	
	Face to Face		w/Tools		Organization		Knowledge		Delay		Problems		Performance		Impact	
Predictor Variable	β	Sig	β	Sig	β	Sig	β	Sig	β	Sig	β	Sig	β	Sig	β	Sig
Number of Time Zones	-0.03	0.645	-0.06	0.292	-0.02	0.683	-0.01	0.849	0.12	0.020		n.s.		n.s.	0.07	0.037
Team Size Core	-0.01	0.445	-0.03	0.026	-0.01	0.567	-0.01	0.478	0.00	0.701	0.03	0.003		n.s.		n.s.
Distance Away (survey)	-0.35	0.000	-0.04	0.521	0.04	0.422	<i>-0.08</i>	<i>0.100</i>	0.07	0.170		n.s.	-0.06	0.176	0.05	0.135
Max Time Zone Span	-0.02	0.426	0.00	0.859	-0.02	0.250	0.02	0.281	0.01	0.566	0.03	0.031	-0.04	0.021	-0.04	0.002
Time Zone Balance (Avg/Max in TZ)	0.40	0.338	-0.25	0.507	0.00	0.998	<i>0.49</i>	<i>0.096</i>	<i>0.57</i>	<i>0.055</i>		n.s.	-0.58	0.024		n.s.
Project Length in Months	0.00	0.930	0.00	0.891	0.00	0.589	0.00	0.643	<i>0.00</i>	<i>0.055</i>		n.s.		n.s.		n.s.
Task Dependency	0.17	0.165	0.31	0.005	0.00	0.982	0.43	0.000	-0.27	0.006	0.35	0.001	0.34	0.000	0.09	0.224
Comm F2F (Comm Mode 1)												n.s.		n.s.		n.s.
Comm Tools (Comm Mode 0, 2-5)											0.27	0.004		n.s.		n.s.
Project Task Organization (PrjCoord 7-9)											-0.74	0.000		n.s.	<i>0.19</i>	<i>0.091</i>
Team Knowledge (PrjAware & PrjKnwl)												n.s.		n.s.	<i>0.18</i>	<i>0.059</i>
Communication Delay											0.33	0.004	0.19	0.116		n.s.
Coordination Problems (PrjCoord 13-21)													-0.64	0.000	-0.20	0.016
R-Square (explained variance)		0.547		0.350		0.061		0.479		0.368		0.533		0.595		0.443
P-Value (predictive power)		0.000		0.000		0.723		0.000		0.000		0.000		0.000		0.000

Figure 1: Research Framework

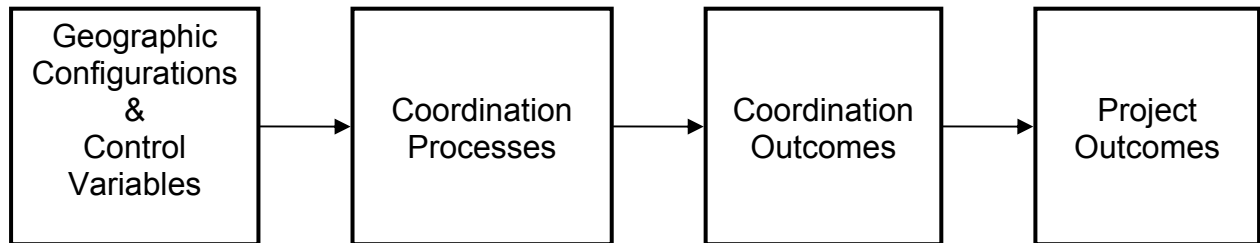


Figure 2: Regression Analysis Results

